

NETWORKS AND COMPLEXITY

Solution 19-1

*This is an example solution from the forthcoming book *Networks and Complexity*.
Find more exercises at <https://github.com/NC-Book/NCB>*

Ex 19.1: Eigenvectors [Lvl. 1]

For the Jacobian below, compute the eigenvalues λ_n , left eigenvectors $\mathbf{w}_n^\dagger$ and right eigenvectors $\mathbf{v}_n$, such that $\mathbf{w}_n^\dagger \mathbf{v}_m = \delta_{n,m}$. Then express the $\mathbf{x} = (0, 1)^\top$ as a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$.

$$\mathbf{J} = \begin{pmatrix} -1 & 3 \\ 0 & -2 \end{pmatrix}$$

Solution

Since the matrix is triangular we can read off the eigenvalues $\lambda_1 = -1$ and $\lambda_2 = -2$ straight away. To find the first right eigenvector we solve

$$\begin{pmatrix} -1 & 3 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = - \begin{pmatrix} a \\ b \end{pmatrix} \quad (1)$$

Note that in this case the equation for b is

$$-2b = b \quad (2)$$

which can only be solved with $b = 0$, so we set $a = 1$ which yields the solution

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \quad (3)$$

For the second right eigenvector we solve

$$\begin{pmatrix} -1 & 3 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = -2 \begin{pmatrix} a \\ b \end{pmatrix} \quad (4)$$

Now $b = 1$ solves the lower row, and we are left with the upper row

$$-a + 3 = -2a \quad (5)$$

$$a = -3 \quad (6)$$

Hence the second right eigenvector is

$$\mathbf{v}_2 = \begin{pmatrix} -3 \\ 1 \end{pmatrix}. \quad (7)$$

To compute the first left eigenvector we need to solve

$$(a, b) \begin{pmatrix} -1 & 3 \\ 0 & -2 \end{pmatrix} = -(a, b) \quad (8)$$

Considering the left and the right columns separately leads to the equations

$$-a + 0b = -a \quad (9)$$

$$3a - 2b = -b \quad (10)$$

The first equation just tells us that we can choose a as we want so we set $a = 1$. Now the remaining equation reads

$$3 - 2b = -b \quad (11)$$

$$b = 3 \quad (12)$$

Hence we can write the eigenvector as

$$\mathbf{w}_1^\dagger = (1, 3) \quad (13)$$

For the second left eigenvector we consider

$$(a, b) \begin{pmatrix} -1 & 3 \\ 0 & -2 \end{pmatrix} = -2(a, b) \quad (14)$$

Now the two columns give us the equations

$$-a = -2a \quad (15)$$

$$3a - 2b = -2b \quad (16)$$

The first equation requires $a = 0$. Then any b will satisfy the second equation so we choose $b = 1$. So before checking the normalization we have

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \mathbf{w}_1^\dagger = (1, 3) \quad (17)$$

$$\mathbf{v}_2 = \begin{pmatrix} -3 \\ 1 \end{pmatrix} \quad \mathbf{w}_2^\dagger = (0, 1) \quad (18)$$

Now let's check the orthonormality condition. We start by checking that left eigenvectors are orthogonal to right eigenvectors that belong to a different eigenvalue

$$\mathbf{w}_1^\dagger \mathbf{v}_2 = (1, 3) \begin{pmatrix} -3 \\ 1 \end{pmatrix} = (-3 + 3) = 0 \quad (19)$$

$$\mathbf{w}_2^\dagger \mathbf{v}_1 = (0, 1) \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 0 + 0 = 0 \quad (20)$$

Now we check the products of eigenvectors belonging to the same eigenvalue. For eigenvalue 1 we find

$$\mathbf{w}_1^\dagger \mathbf{v}_1 = (1, 3) \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1 \quad (21)$$

which is perfect. There is still one product remaining,

$$\mathbf{w}_2^\dagger \mathbf{v}_2 = (0, 1) \begin{pmatrix} -3 \\ 1 \end{pmatrix} = 1, \quad (22)$$

which is also correctly normalized already. This shows that we have found the vectors we wanted. Note however, that this solution is not unique. In this case choosing these vectors is

quite compelling, but if you double the length of say $\mathbf{v}_1$ and halve the length of $\mathbf{w}_n^\dagger$ you will still have vectors that meet the conditions, and consequently you will also find slightly different answers to the last part of this exercise.

We now want to find c_1 and c_2 such that

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (23)$$

From the chapter we know that we can find the coefficients by multiplying the corresponding left eigenvectors

$$c_1 = \mathbf{w}_1^\dagger \begin{pmatrix} 0 \\ 1 \end{pmatrix} = (1, 3) \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 3 \quad (24)$$

$$c_2 = \mathbf{w}_2^\dagger \begin{pmatrix} 0 \\ 1 \end{pmatrix} = (0, 1) \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 1 \quad (25)$$

Hence we can decompose the vector as

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} = 3\mathbf{v}_1 + 1\mathbf{v}_2 = 3 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 1 \begin{pmatrix} -3 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (26)$$